

PPP Worked Walkthrough

Six-page public entry draft: a facial-recognition alert that points to the wrong person

Scope. This example uses fictional names and a simplified case structure. It reflects documented real-world failure patterns: an uncertain AI match acquires artificial authority as it moves through police and court paperwork.

Situation. A crime-scene still image is entered into a police facial-recognition system. The system returns: “Possible match: Jordan Reed — 93% similarity.” Jordan is not the person who committed the crime.

Question. The point of the walkthrough is not to ask whether police may ever investigate a lead. It is to show how the same alert can produce two very different outcomes depending on whether uncertainty is preserved or silently promoted into guilt.

Diagram. Quick route map

Scene image → model output → database candidate



PAL: what was actually observed and what remains unresolved?



PECAN: is the route authorized, bounded, and reviewable?



PEA: what may proceed, pause, narrow, or be refused?



Accountable human decision



Seed release: explain the result without hiding uncertainty or consuming agency.

Split. Two path comparison

- Failure path: “possible match” becomes “the system identified Jordan,” then becomes suspicion, warrant language, arrest, and prosecution pressure.
- Governed path: the alert remains a candidate lead; independent corroboration is required; arrest, search, or accusation are blocked until new evidence appears.

Why this matters. Core lesson: AI does not imprison people by itself. People and institutions allow an uncertain output to cross successive authority boundaries without forcing it to remain uncertain.

Page 2 — PAL

What was actually established?

Task. PAL separates the event into claim layers instead of letting a smooth sentence hide the jump from image to guilt.

Known. What is known:

- A camera captured a low-quality or partial image of a face near the crime scene.
- A facial-recognition model extracted features and searched a comparison database.
- The model returned one or more ranked candidates with a similarity score.
- Jordan Reed appears somewhere in the searched database and was returned as a candidate.

Unresolved. What is not yet known:

- Whether the image is clear enough for dependable identity work.
- Whether the ranking threshold, demographic performance, or database quality distorted the result.
- Whether Jordan was at the scene, resembles the actual person, or has any connection to the event.
- Whether the current lead is one of many plausible candidates or merely the top candidate in a weak search.

Drift. Dangerous compression:

- Valid statement: “The system produced a candidate match.”
- Invalid promotion: “The system identified Jordan.”
- Further invalid promotion: “Jordan likely committed the crime.”

PAL description. PAL output for the next layer:

- Object: one machine-generated candidate association tied to a specific image and search configuration.
- Trace: image source, timestamp, model, database, threshold, analyst steps, and confidence bounds.
- Residual: identity unresolved; guilt unresolved; downstream action must not treat the match as completed fact.

Result. PAL therefore protects the first boundary: similarity is not identity; identity is not culpability; and none of those automatically authorizes police action.

Page 3 — PECAN (Failure Path)

How the alert wrongly hardens into accusation

Pattern. In the failure path, the facial-recognition result is entered into paperwork as though it were already identification evidence. The packet still looks procedural, but the route has silently crossed into unjustified consequence.

Field	Failure-path fill
Purpose	Identify and arrest offender immediately.
Action	Seek warrant, detain Jordan, and pressure follow-up identification.
Evidence	Recorded as “AI identification,” with limitations minimized.
Affected set	Victim and public emphasized; Jordan weakly represented.
Authority	Police authority assumed to cover reliance on the output.
Risk	False-match harm discounted; urgency dominates.
Containment	Arrest now, investigate later.
Outgoing trace	“AI-identified suspect” copied into reports, lineups, warrants, and prosecutor files.

Defect. Why this fails:

- One source is repeated across many documents and mistaken for many independent sources.
- The packet promotes a candidate lead into detention, search, or prosecution pressure without new corroboration.
- Later reviewers inherit the earlier mislabeling and treat the route as already validated.

Verdict. PECAN verdict on this route: BLOCKED PROMOTION. The route is not fit for arrest, charging, or public accusation on the basis of the match alone.

Page 4 — PECAN (Governed Path)

How the same alert becomes a bounded investigative lead

Pattern. In the governed path, the same AI output is routed honestly. The packet may still support investigation, but only inside a narrow authority envelope.

Field	Governed-path fill
Purpose	Investigate a possible lead while protecting public safety and the candidate's standing.
Action	Perform bounded corroboration before arrest, charge, or public accusation.
Evidence	Label the result as an unverified facial-recognition candidate match.
Provenance	Preserve image quality, model, threshold, database, and analyst steps.
Affected set	Jordan, family, victim, officers, and public all remain visible.
Authority	Investigative follow-up allowed; arrest/search not granted by the match alone.
Containment	Require independent corroboration; do not contaminate witness procedures.
Remedy/Governance	If contradicted, close the lead, correct the record, and preserve review/audit.

Result. PECAN output:

- PECAN-VALID only for a bounded investigative lead.
- Not valid for automatic identity claims, arrest, search, charge, or public accusation.
- If no independent evidence appears, the route must expire or be closed rather than kept alive by repetition.

Meaning. Core protection: a complete packet means the proposal is reviewable, not true, ethical, or executable. The protocol preserves uncertainty instead of laundering it into authority.

Page 5 — PEA

What may proceed, pause, narrow, or be refused?

Frame. PEA evaluates the same situation under hard floors: equal standing, non-substitution of remedy, contestability, reversibility where possible, and the burden created by serious error.

Failure path	Governed path
Likely judgment: REFUSE or RETURN TO PECAN Reasons: • unresolved identity promoted into accusation • one source repeated as if independent • detention pressure before proportionate corroboration • weak contest route and slow remedy • foreseeable harm to liberty, work, family, and reputation	Likely judgment: PASS WITH CONDITIONS for continued investigation Conditions: • obtain independent corroboration • disclose FR role and limits internally • prevent contamination of lineups or witness memory • no arrest on the match alone • expire/close the lead when contradicted • preserve a rapid contest and correction route

Difference. PEA does not say “never investigate.” It says public safety may justify careful investigation, but uncertainty may not be converted into guilt merely because a machine score looks impressive or has been repeated in official language.

Conclusion. Candidate ethical conclusion: the alert may support a narrowly bounded follow-up, but not coercive action unless independent evidence changes the case.

Page 6 — Accountable Human Boundary and Seed Release

Who decides, who carries responsibility, and how the result is explained

Accountability. No single actor may hide behind the phrase “the system said so.” Responsibility remains layered and human.

- Vendor or developer: model limitations, disclosed performance, auditability, and known error modes.
- Police department: deployment rules, thresholds, training, supervisory review, retention, and remedy policy.
- Analyst or officer: honest characterization of the match, bounded interpretation, and proportionate action in the actual situation.
- Judge or prosecutor: independent review; repeated paperwork is not independent evidence.
- Institution as a whole: correction, notice, audit, and repair if a false association harmed someone.

Seed release. Driver-facing / person-facing release

“A facial-recognition search returned your image as a possible candidate. The result was not an identification, and independent review showed that you were not involved. The lead has been closed and marked as a false association. Here is how to obtain the record, request correction, and report any continuing effect.”

Why Seed. Why this is a Seed release:

- It gives the person the real reason for the contact.
- It names uncertainty instead of burying it.
- It leaves a route for correction, challenge, and future repair.
- It stops when the case stops instead of consuming the person with endless procedural language.

Takeaway. Final takeaway: the humane system does not treat a model output as guilt, and it does not use complexity to hide who was responsible for crossing that boundary.